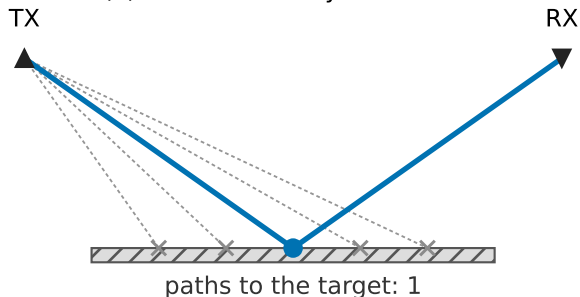


One ray engine (Mitsuba 3.8.0, Dr.Jit 1.3.1) shoots rays and finds triangle hits

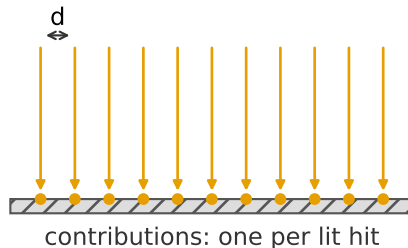
(a) Path solver: rays are scouts



N probe rays on a quasi-uniform lattice
N: 1e+04 to 4e+06 (400x), amplitude spread 0.0 dB
all of them report one surface sequence,
so the first is kept and the rest discarded
the image method then places the exact point
sb_candidate_generator.py:484-498

field update at that one point
a = pattern x Fresnel Jones
x 1/(s' + s) x lambda/4pi
args: to_world, ki_local, ko_local,
reflection, r_te, r_tm, t_te, t_tm
no area, no curvature, no triangle size
radio_material.py:853-863

(b) PO integral: hits are the answer



parallel rays on a uniform grid
over the aperture seen by the radar
grid spacing $d = \lambda / 16$ in production,
so one ray stands for the area d^2
the hit point and its phase are both kept
src/rcs_sbr.py

coherent sum over every lit point
$$E = \sum |\Gamma_i| \exp(j 2k p_i \cdot u) \times d^2$$
$$\sigma = (4 \pi / \lambda^2) |E|^2$$

more lit area means more terms
self-shadowing is free: first hits only
size enters here, and only here